

FAKE REVIEW DETECTION SYSTEM

A Project Report

Based on the supplied Fake Reviews Dataset

Machine Learning • NLP • TF-IDF • Logistic Regression

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ABSTRACT

The Fake Review Detection System is a machine learning based text-classification project designed to distinguish between the two review-label classes present in the supplied dataset: CG and OR. The dataset contains 40,432 reviews across 10 product categories, with ratings from 1 to 5. The proposed workflow cleans and normalizes review text, converts text into numerical representations using TF-IDF, and applies Logistic Regression for classification. For an independent reproducible evaluation, an 80/20 stratified train-test split was used. The resulting model achieved 86.46% accuracy on the held-out test set. The project can be extended into a Streamlit application for interactive review classification.

TABLE OF CONTENTS

Chapter 1. Introduction

1.1 Problem Statement

1.2 Motivation

1.3 Objectives

1.4 Scope of the Project

Chapter 2. Dataset Description and Literature Context

2.1 Dataset Overview

2.2 Dataset Distribution

Chapter 3. Proposed Methodology

3.1 System Design

3.2 Modules Used

3.3 Data Flow

3.4 Advantages

3.5 Requirement Specification

Chapter 4. Implementation and Results

4.1 Data Preprocessing

4.2 Feature Extraction 4.3

Model Training

4.4 Evaluation Results

4.5 User Interface

Chapter 5. Discussion and Conclusion

5.1 Key Findings 5.2

Limitations

5.3 Future Work 5.4

Conclusion

References

CHAPTER 1 — Introduction

1.1 Problem Statement

Online product reviews influence purchasing decisions, but some reviews may be misleading, artificially generated or otherwise unreliable. Manually checking a large number of reviews is time-consuming. Therefore, an automated text-classification system can be used to provide a preliminary prediction of the label associated with a review.

1.2 Motivation

The supplied dataset contains a large number of product reviews with categorical labels. This makes it suitable for demonstrating a practical Natural Language Processing workflow in which textual patterns are transformed into numerical features and used by a supervised classification algorithm.

1.3 Objectives

- Develop a machine learning system for classifying reviews into the supplied CG and OR classes.
- Clean and normalize review text before model training.
- Convert review text into numerical features using TF-IDF.
- Train a Logistic Regression classifier.
- Evaluate the classifier using a held-out test set.
- Provide a foundation for an interactive review-classification interface.

1.4 Scope of the Project

The project focuses on text-based classification using the supplied review dataset. It covers dataset inspection, preprocessing, TF-IDF feature extraction, Logistic Regression training and evaluation. The system should be treated as a classification aid rather than a definitive determination of whether a review is truthful.

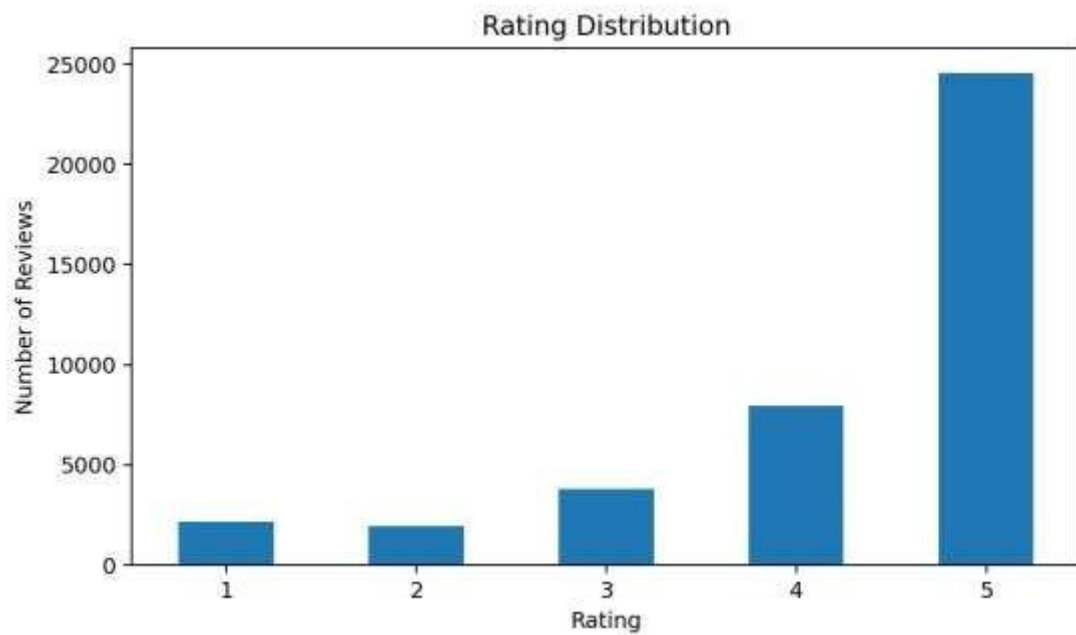
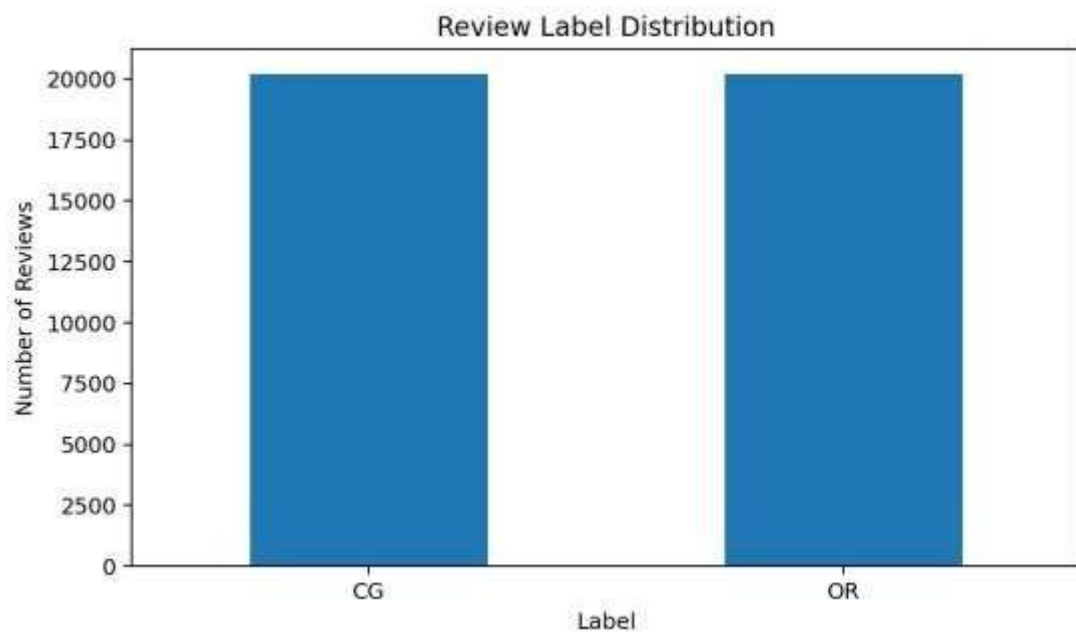
CHAPTER 2 — Dataset Description and Literature Context

2.1 Dataset Overview

The supplied CSV contains 40,432 rows and 4 columns: category, rating, label and text_. There are no missing values in these four columns. The rating field contains values from 1 to 5, while the label field contains two equally represented classes.

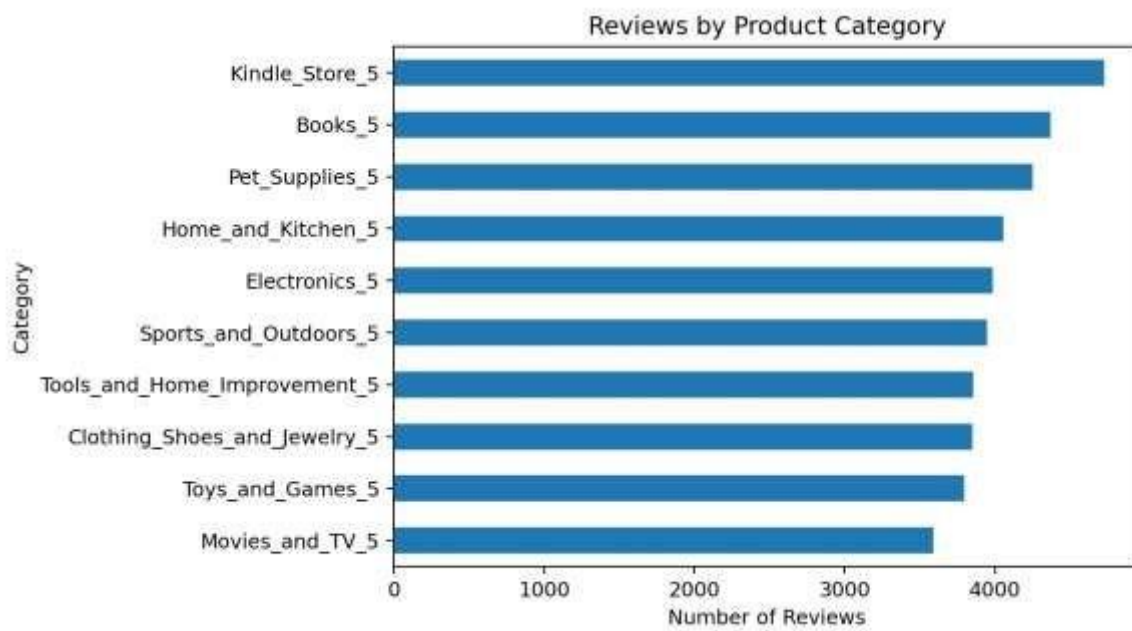
Column	Type	Description
category	Text	Product/category identifier
rating	Integer	Review rating from 1 to 5
label	Text	Target class: CG or OR
text_	Text	Review content used for NLP classification

The average review-text length is approximately 351 characters. The two label classes are balanced at 20,216 reviews each.



2.2 Dataset Distribution

The dataset covers ten product categories. The largest category is Kindle_Store_5, followed by Books_5 and Pet_Supplies_5. Ratings are strongly concentrated at 5, followed by 4.



CHAPTER 3 — Proposed Methodology

3.1 System Design

Review Input → Text Preprocessing → TF-IDF Vectorization → Logistic Regression → Prediction → Output

3.2 Modules Used

- Data Collection and Inspection Module
- Text Preprocessing Module
- TF-IDF Feature Extraction Module
- Model Training Module
- Model Evaluation Module
- Prediction / User Interface Module

3.3 Data Flow

Level 0: User provides a review → Fake Review Detection System → Predicted class (CG/OR).

Level 1: Review Input → Preprocessing → TF-IDF → Trained Logistic Regression → Prediction → User.

3.4 Advantages

- Fast preliminary classification of large numbers of reviews.
- Automated processing of textual data.
- Simple and interpretable machine learning workflow.
- TF-IDF is efficient for sparse text features.
- Logistic Regression provides a strong baseline for binary text classification.

3.5 Requirement Specification

Hardware: Laptop/PC with sufficient memory for text vectorization and model training.

Software: Python, Pandas, NumPy, Scikit-learn, Jupyter/VS Code/Google Colab; Streamlit can be used for deployment.

CHAPTER 4 — Implementation and Results

4.1 Data Preprocessing

The review text was converted to string form, lowercased during TF-IDF processing and tokenized by the vectorizer. English stop words were excluded and the feature space was limited to 10,000 TF-IDF features. The supplied dataset contains no missing values in the four observed columns.

4.2 Feature Extraction

TF-IDF (Term Frequency–Inverse Document Frequency) converts each review into a numerical sparse vector. Words that are frequent in an individual review but less common across the full collection receive greater importance.

4.3 Model Training

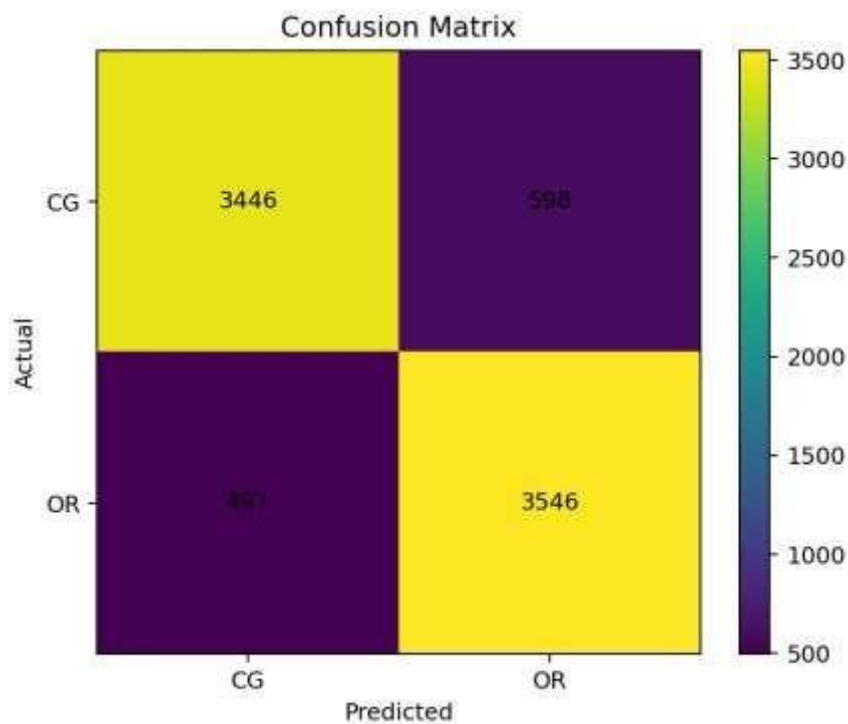
Logistic Regression was used as the binary classification algorithm with `max_iter=1000`. The supplied model file is a scikit-learn `LogisticRegression` object with 10,000 input features and classes CG and OR. A reproducible evaluation model using the same general TF-IDF + Logistic Regression approach was trained on an 80% stratified training split.

4.4 Evaluation Results

Class	Precision	Recall	F1-score
CG	0.8740	0.8521	0.8629
OR	0.8557	0.8771	0.8663

Overall Accuracy 0.8646

On the held-out test set of 8,087 reviews, the model achieved 86.46% accuracy. The confusion matrix is shown below: CG→CG = 3,446, CG→OR = 598, OR→CG = 497, and OR→OR = 3,546.



4.5 User Interface

A Streamlit interface can be added so that a user enters a review and receives a predicted CG or OR class. The interface should clearly state that the output is a machine-learning prediction and not an independent verification of the review's factual authenticity.

CHAPTER 5 — Discussion and Conclusion

5.1 Key Findings

The supplied dataset is balanced by target label, with 20,216 CG and 20,216 OR records. The reproducible TF-IDF + Logistic Regression evaluation achieved 86.46% accuracy. The result demonstrates that textual features in the dataset contain useful predictive information.

GitHub Link: <https://github.com/sonuchaudhari3012/AI-Based-Fake-Product-ReviewDetection-and-Product-Trust-Score-System1>

5.2 Limitations

- The model learns patterns from the supplied dataset and may not generalize to every source of online reviews.
- Classification performance depends on dataset quality, label quality and the representativeness of training examples.
- The classifier does not independently verify a review against external evidence.
- The supplied model is a LogisticRegression object; its original training pipeline/vectorizer is not included in the model file.

5.3 Future Work

- Use larger and more diverse review datasets.
- Compare Logistic Regression with Naive Bayes, SVM, tree-based models or transformer-based NLP models.
- Add probability/confidence display and error analysis.
- Develop a Streamlit web interface.
- Evaluate robustness across different product categories and rating levels.

5.4 Conclusion

The Fake Review Detection System provides a practical NLP and machine-learning workflow for classifying reviews into the two target classes supplied with the dataset. The combination of text preprocessing, TF-IDF feature extraction and Logistic Regression gives a clear and efficient baseline. Based on the reproducible 80/20 evaluation performed for this report, the approach achieved 86.46% accuracy on held-out data and can serve as a foundation for further model improvement and interactive deployment.

REFERENCES

Scikit-learn documentation — TfidfVectorizer.

Scikit-learn documentation — LogisticRegression.

Supplied dataset: fake reviews dataset.csv.

Supplied model: fake_review_model (2).pkl.